

Customer Churn Prediction

Machine Learning Project Report
Classification, Model Evaluation, and Dataset Shift Analysis

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Year: 2026

Project objective

Build and evaluate customer churn classifiers using the provided training dataset, then assess whether their performance generalizes to the independent test dataset.

Dataset

Kaggle Customer Churn Dataset — Muhammad Shahid Azeem

Executive Summary

This project developed customer churn classification models using a large customer dataset containing demographic, account, usage, support, payment, contract, spending, and interaction variables. The target variable, Churn, indicates whether a customer left the service.

Exploratory analysis identified strong relationships between churn and variables such as Support Calls, Payment Delay, Total Spend, Age, and Contract Length. Logistic Regression was used as an interpretable baseline, followed by a Random Forest model capable of learning nonlinear interactions.

Within the supplied training dataset, Random Forest produced near-perfect validation and cross-validation performance. However, evaluation on the separately supplied test dataset produced a severe performance drop. Further investigation showed substantial changes in both feature distributions and feature-to-target relationships between the training and test files. The findings are consistent with significant dataset/distribution shift.

The main conclusion is therefore not that a production-ready churn model was achieved. Rather, the project demonstrates why independent holdout evaluation and dataset-shift analysis are essential: excellent validation metrics within one data distribution do not guarantee reliable performance on a different population.

1. Business Problem

Customer churn prediction is a binary classification problem. The goal is to estimate whether a customer is likely to churn (1) or remain active (0) from information available about that customer. A reliable model could support retention efforts by helping a business identify customers who may require proactive support or targeted intervention.

In this project, model quality was assessed not only by accuracy, but also by Precision, Recall, F1-score, ROC-AUC, confusion matrices, prediction distributions, cross-validation stability, and performance on an independent test file.

2. Dataset Overview

The supplied training data contained 440,833 rows before cleaning and 12 columns. One fully empty row was identified and removed, leaving 440,832 usable observations. No duplicate rows were detected. The independent test dataset contained 64,374 observations and no missing values.

Variable	Role / Meaning
CustomerID	Identifier; excluded from model features
Age	Customer age
Gender	Categorical customer gender
Tenure	Length of customer relationship
Usage Frequency	Service usage frequency
Support Calls	Number of customer support calls
Payment Delay	Payment delay measure
Subscription Type	Basic, Premium, or Standard
Contract Length	Monthly, Quarterly, or Annual
Total Spend	Customer spending
Last Interaction	Recency/interaction measure

Churn	Binary target: 0 = No Churn, 1 = Churn
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CustomerID was removed because it functions as an identifier rather than a meaningful predictive characteristic. The remaining ten variables were used as model inputs.

3. Exploratory Data Analysis

3.1 Target Distribution

The cleaned training dataset was moderately balanced: approximately 56.71% of observations were churned customers and 43.29% were non-churned customers. Because neither class was extremely rare, the project did not require aggressive class-balancing techniques at this stage.

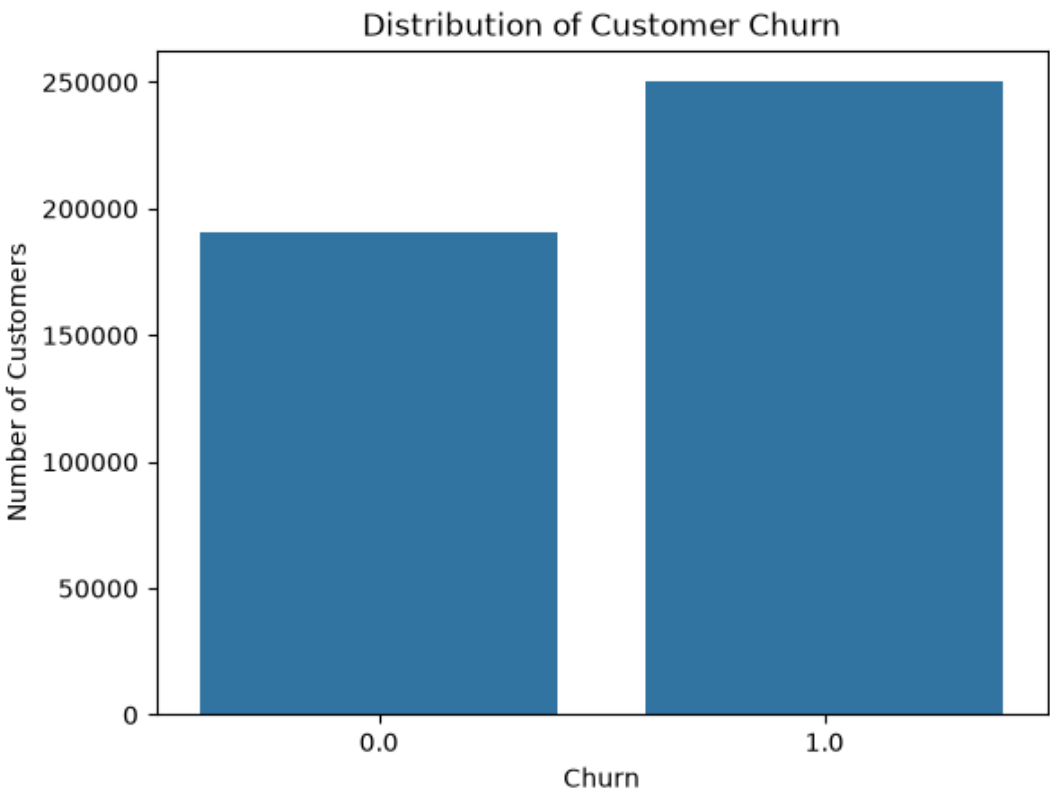


Figure 1. Distribution of the churn target in the cleaned training dataset.

3.2 Numerical Relationships

Feature	Mean — No Churn	Mean — Churn	Observed Pattern
Age	36.26	41.75	Churned customers were older on average
Support Calls	1.59	5.14	Large increase among churned customers
Payment Delay	10.02	15.22	Higher among churned customers
Total Spend	749.95	541.29	Lower among churned customers
Last Interaction	13.01	15.60	Higher among churned customers

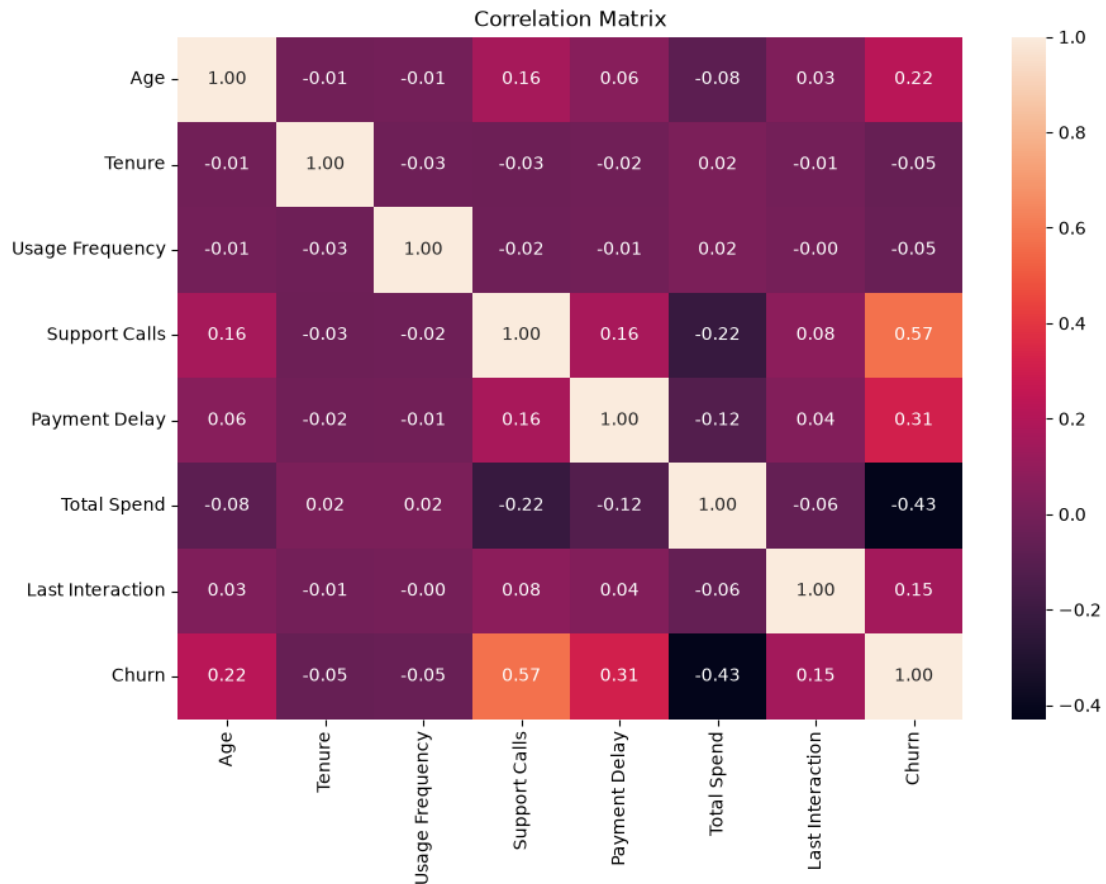


Figure 4. Correlation matrix for numerical variables in the training dataset.

The correlation analysis supported these observations. Churn had its strongest positive numerical correlation with Support Calls (approximately 0.57), followed by Payment Delay (approximately 0.31) and Age (approximately 0.22). Total Spend had a notable negative correlation with Churn (approximately -0.43). Correlation is descriptive rather than causal; these values do not establish that any feature causes churn.

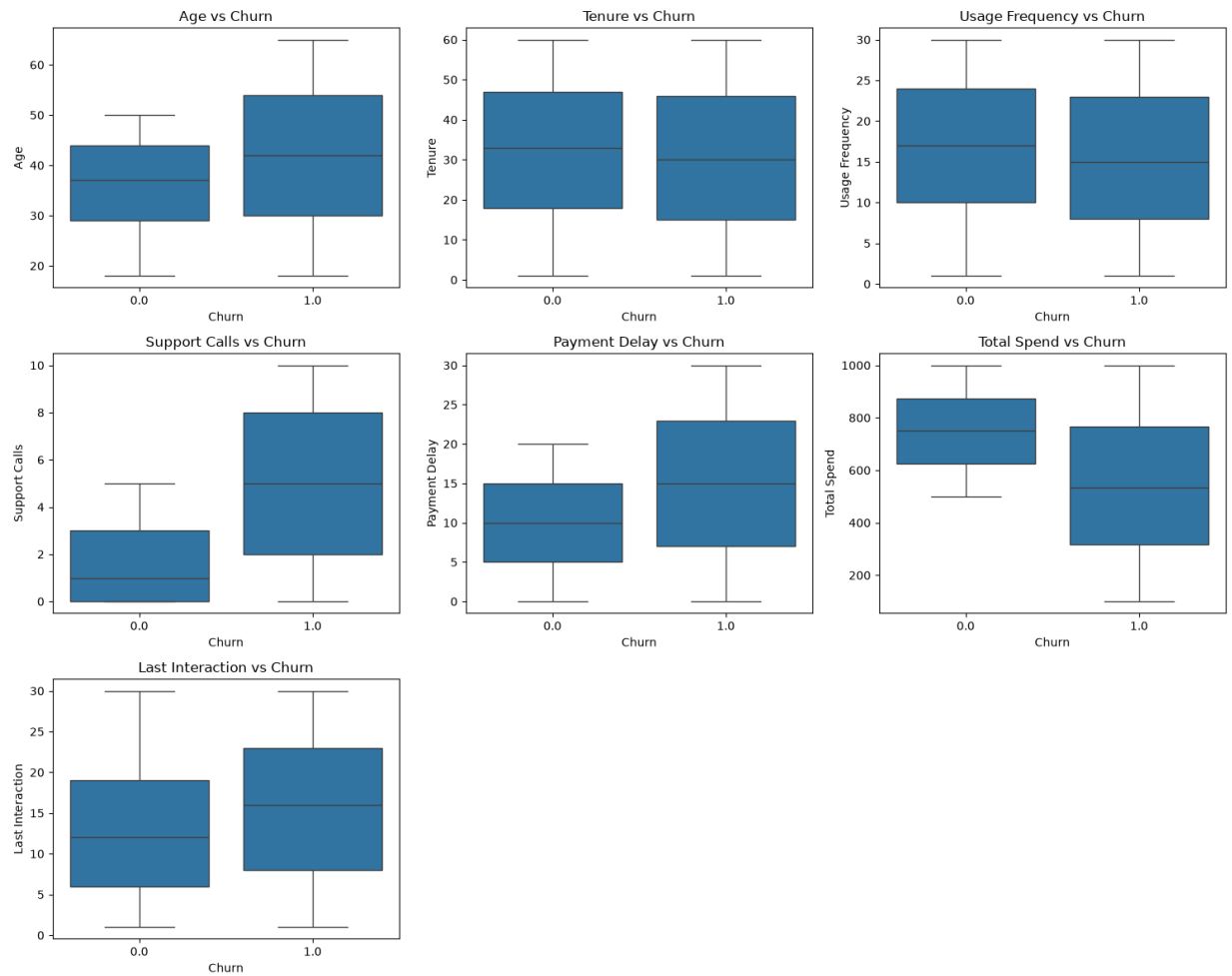


Figure 2. Numerical feature distributions by churn status.

3.3 Categorical Relationships

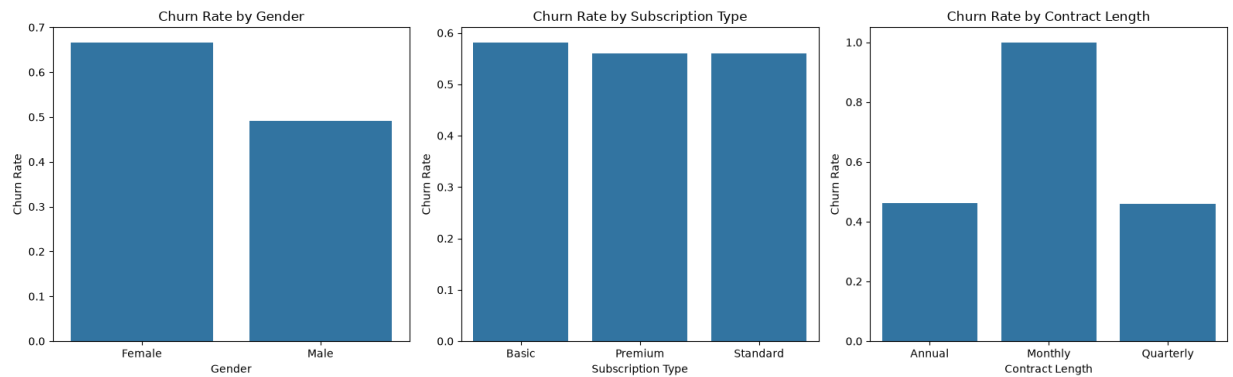


Figure 3. Training-set churn rates by gender, subscription type, and contract length.

Contract Length showed an especially strong pattern in the training data: all customers with a Monthly contract were labelled as churned. Annual and Quarterly contracts each had churn rates of roughly 46%. Female customers also had a higher observed churn rate than male customers in the training data. Subscription Type showed comparatively small differences.

The perfect association between Monthly contracts and Churn was treated as a warning sign rather than as proof of a general business rule. A model can exploit such a pattern extremely effectively within the training distribution even when that pattern does not hold in a separate population.

4. Data Preparation

The target variable Churn was separated from the input features. The training data was split into 80% training and 20% validation data using stratification so that both subsets preserved the original churn distribution.

- Numerical features were standardized with StandardScaler.
- Categorical features were transformed with OneHotEncoder using `handle_unknown='ignore'`.
- Preprocessing and estimation were combined in scikit-learn Pipelines to ensure consistent transformations.
- The independent test file was not used for model fitting or hyperparameter selection.

This pipeline-based approach also reduces the risk of preprocessing inconsistencies between training, validation, and test data.

5. Models

5.1 Logistic Regression

Logistic Regression was selected as the baseline classifier because it is efficient, interpretable, and provides a useful linear benchmark. On the validation set, it achieved approximately 89.34% accuracy. Its ROC-AUC was approximately 0.959, showing strong ranking performance within the training-data distribution.

5.2 Random Forest

Random Forest was then used to capture nonlinear relationships and feature interactions. Its validation accuracy reached approximately 99.93%, with training accuracy of 100.00%. The training-validation accuracy difference was only about 0.07 percentage points, so a single validation split alone did not show the usual large gap associated with obvious overfitting.

6. Random Forest Diagnostics

6.1 Confusion Matrix

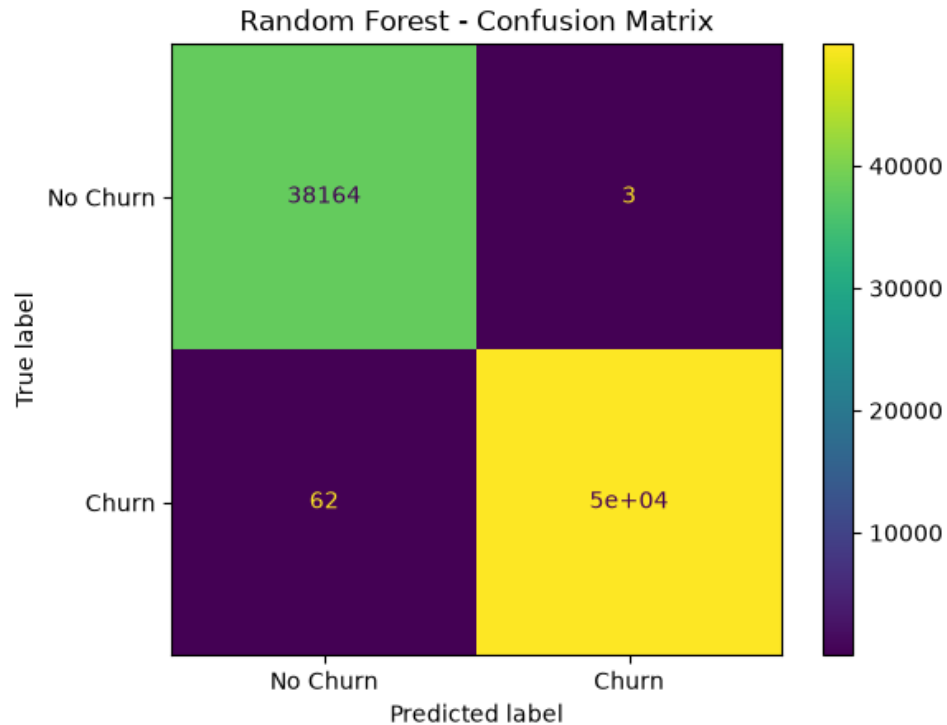


Figure 5. Random Forest confusion matrix on the internal validation set.

On the validation set, Random Forest produced 38,164 true negatives, 3 false positives, 62 false negatives, and approximately 50,000 true positives. This result was unusually strong and motivated additional robustness checks rather than immediate acceptance.

6.2 Feature Importance

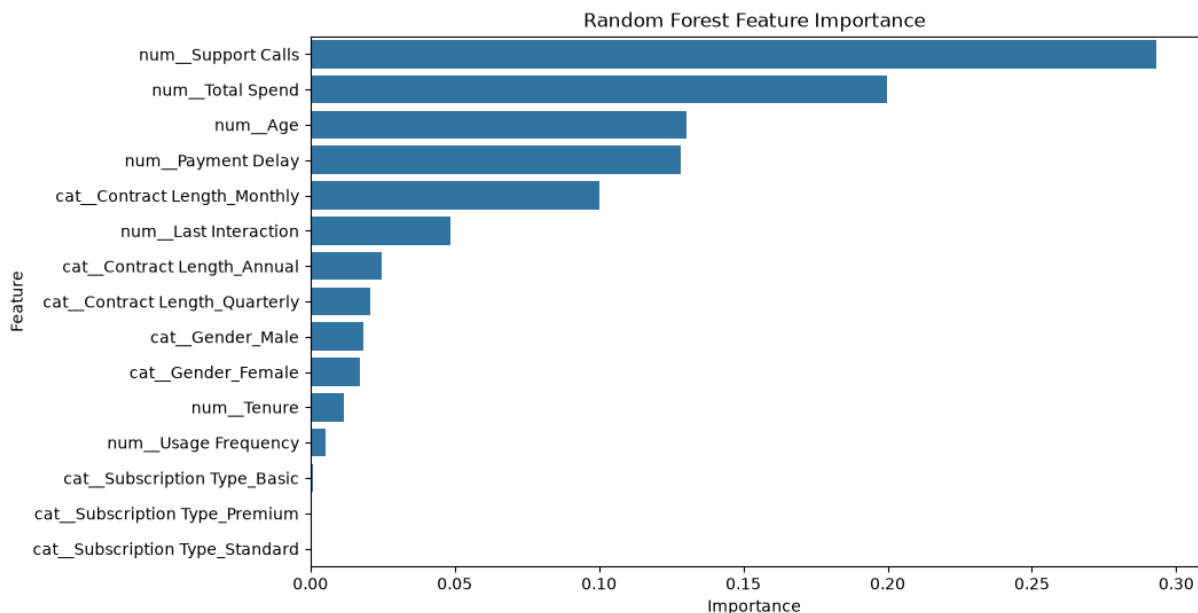


Figure 6. Random Forest feature importance on the training/validation workflow.

Random Forest feature importance indicated that Support Calls was the most influential feature, followed by Total Spend, Age, Payment Delay, and the Monthly Contract Length indicator. Subscription Type contributed very little to the fitted model.

Approximate Rank	Feature	Approximate Importance
1	Support Calls	~0.29
2	Total Spend	~0.20
3	Age	~0.13
4	Payment Delay	~0.13
5	Contract Length — Monthly	~0.10

Feature importance indicates how useful a variable was for the Random Forest's splits; it should not be interpreted as causal effect.

6.3 Contract-Length Sensitivity Check

Because Monthly contracts were perfectly associated with churn in the training data, a second Random Forest was evaluated without Contract Length. Validation accuracy remained very high at approximately 98.80%, with ROC-AUC around 0.9896. Therefore, the near-perfect validation performance was not explained by the Monthly contract feature alone.

6.4 Five-Fold Cross-Validation

A stratified five-fold cross-validation experiment was performed to test whether the Random Forest result depended on a fortunate train/validation split. The scores were extremely stable.

Fold	Accuracy
1	99.92%
2	99.92%
3	99.91%
4	99.93%
5	99.94%
Mean	99.92%
Standard deviation	0.0072%

Mean Precision was approximately 1.0000, Mean Recall 0.9987, Mean F1-score 0.9993, and Mean ROC-AUC 1.0000. This confirmed that the model was highly stable when resampling observations from the same training-data distribution.

7. Independent Test Evaluation

The decisive evaluation used the separately supplied test file containing 64,374 customers. This file included the true Churn labels, allowing direct measurement of out-of-sample performance.

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	57.12%	52.51%	99.06%	68.64%	0.6894
Random Forest	50.41%	48.85%	99.87%	65.61%	0.6123

Both models suffered a substantial decline compared with their validation results. The very high Recall values were misleading when viewed alone: Logistic Regression predicted Churn for approximately 89.36% of test customers, while Random Forest predicted Churn for approximately 96.84%. In reality, only about 47.37% of the test customers were churned.

8. Dataset Shift Investigation

8.1 Target Distribution

Dataset	No Churn	Churn
Training	43.29%	56.71%
Independent Test	52.63%	47.37%

The target prevalence changed between the two datasets, but this difference alone was not large enough to explain why the models predicted Churn for nearly all test customers.

8.2 Feature Distribution Changes

Feature	Training Mean	Test Mean	Direction
Age	39.38	41.97	Higher in test
Support Calls	3.61	5.40	Substantially higher in test
Payment Delay	12.97	17.13	Substantially higher in test
Total Spend	631.89	541.02	Substantially lower in test
Last Interaction	14.48	15.50	Higher in test
Tenure	31.24	31.99	Similar
Usage Frequency	15.82	15.08	Similar/slightly lower

Several shifts occurred in the same direction that the training data associated with churn: more support calls, greater payment delay, and lower total spending. Consequently, many ordinary test customers resembled churned customers from the training distribution.

8.3 Contract-Length Shift

Contract Length	Training	Test
Annual	40.13%	33.26%
Quarterly	40.06%	32.36%
Monthly	19.80%	34.38%

Monthly contracts increased from approximately 19.8% of the training data to 34.4% of the test data. Because every Monthly customer in the training data was labelled as churned, this change further pushed the fitted models toward Churn predictions.

8.4 Changes Within Churn Classes

Dataset / Class	Support Calls	Payment Delay	Total Spend	Age	Last Interaction
Train — No Churn	1.59	10.02	749.95	36.26	13.01
Train — Churn	5.14	15.22	541.29	41.75	15.60
Test — No Churn	4.50	12.45	560.54	41.13	15.52
Test — Churn	6.40	22.33	519.34	42.90	15.47

This comparison is particularly important. For example, non-churned customers in the test file averaged 4.50 support calls, much closer to churned customers in the training file (5.14) than to non-churned training customers (1.59). Similarly, average Total Spend for non-churned test customers (560.54) was close to the churned training average (541.29). This shows that not only the marginal feature distributions, but also some feature-to-target relationships differed across the supplied files.

9. Interpretation

The evidence is consistent with significant dataset shift between the supplied training and test files. The models learned patterns that were highly predictive inside the training population, and cross-validation confirmed that those patterns were stable when sampling from that same population. The independent test population, however, followed materially different distributions and relationships, causing both models to generalize poorly.

This distinction is important: cross-validation protects against dependence on one random split, but it does not guarantee robustness to a different data-generating process. A model can therefore achieve excellent cross-validation metrics and still fail after deployment if production data differ from the development data.

10. Limitations

- The original data-generation process, collection periods, population definitions, and exact method used to create the supplied training and test files are not established within this project.
- Because the independent test set was used for final diagnosis, repeatedly tuning models to improve this test score would contaminate its role as an unbiased holdout set.
- Observed associations such as high support-call volume and churn should not be interpreted as causal relationships.
- Near-perfect in-distribution validation metrics should not be presented as evidence of production performance.

11. Recommendations

- Obtain additional representative data sampled from the population and time period in which the model will actually be used.
- Document the data-generation and train/test splitting process, including whether the files represent different periods, customer cohorts, or systems.
- Use time-aware or group-aware validation if the operational setting contains temporal or customer-group shifts.
- Monitor feature distributions and prediction rates after deployment, with explicit drift detection for important variables such as Support Calls, Payment Delay, Total Spend, and Contract Length.
- Do not deploy the current models as production churn classifiers based solely on the supplied datasets.

12. Project Outcome

The project successfully implemented an end-to-end classification workflow: data inspection and cleaning, exploratory analysis, feature preprocessing, baseline and nonlinear models, confusion-matrix analysis,

probability-based evaluation, feature importance, cross-validation, independent holdout testing, and dataset-shift investigation.

The most important result was the identification of a major generalization problem that would have been hidden if evaluation had stopped at validation accuracy. This demonstrates a practical machine-learning lesson: reliable model development requires testing assumptions about the data distribution, not simply optimizing a headline metric.

13. Reproducibility and Technology Stack

Component	Technology
Language	Python
Data manipulation	pandas, NumPy
Visualization	Matplotlib, Seaborn
Machine learning	scikit-learn
Preprocessing	StandardScaler, OneHotEncoder, ColumnTransformer
Models	Logistic Regression, Random Forest
Evaluation	Accuracy, Precision, Recall, F1-score, ROC-AUC, Confusion Matrix, 5-Fold CV
Development environment	Visual Studio Code / Jupyter Notebook

Appendix A — Key Results at a Glance

Experiment	Result
Training records after cleaning	440,832
Independent test records	64,374
Training churn rate	56.71%
Random Forest validation accuracy	99.93%
Random Forest 5-fold CV mean accuracy	99.92%
Random Forest CV accuracy standard deviation	0.0072%
Random Forest independent test accuracy	50.41%
Logistic Regression independent test accuracy	57.12%
Random Forest predicted Churn on test	96.84%
Logistic Regression predicted Churn on test	89.36%
Actual Churn in test	47.37%

Appendix B — Dataset Reference

Dataset: Customer Churn Dataset, published on Kaggle by Muhammad Shahid Azeem.

Source URL: <https://www.kaggle.com/datasets/muhammadshahidazeem/customer-churn-dataset>

Note: Community discussions around this dataset report similar discrepancies between validation performance on the supplied training file and performance on the separately supplied test file. The conclusions in this report, however, are based on the project's own measured results and direct comparison of the two provided datasets.